

MTH 418a: Inference-I
Assignment No. 6: Unbiased Estimation, UMVUE,
Lehmann-Scheffe Theorem

1. Let $X_1, \dots, X_n$ be a random sample from p.m.f./p.d.f. $f_\theta, \theta \in \Theta$. In each of the following cases, find the UMVUE of $\psi(\theta)$ if it exists.
 - (i) $X_1 \sim N(\theta, \sigma_0^2), \Theta = \mathbb{R}, \psi(\theta) = \theta^r, r = 1, 2, 3, 4, \sigma_0$ is a known positive constant;
 - (ii) $X_1 \sim N(\mu_0, \theta^2), \Theta = (0, \infty), \psi(\theta) = \theta^r, r > -n, \mu_0$ is a known real constant;
 - (iii) $X_1 \sim \text{Exp}(\mu_0, \theta), \Theta = (0, \infty), \psi(\theta) = \theta^r, r > -n, \mu_0$ is a known real constant;
 - (iv) $X_1 \sim \text{Exp}(\mu_0, \theta), \Theta = (0, \infty), \psi(\theta) = P_\theta(X_1 \leq t_0)$, where μ_0 and t_0 are fixed real constants;
 - (v) $X_1 \sim \text{Exp}(\theta, \sigma_0), \Theta = \mathbb{R}, \psi_r(\theta) = \theta^r, r = 1, 2, 3, 4$, where σ_0 is a fixed positive constant;
 - (vi) $X_1 \sim U(\mu_0 - \theta, \mu_0 + \theta), \Theta = (0, \infty), \psi(\theta) = \theta^r, r > -n, \mu_0$ is a known real constant;
 - (vii) $f_\theta(x) = \alpha_0 \frac{x^{\alpha_0-1}}{\theta^{\alpha_0}} I(0 < x < \theta), \Theta = (0, \infty), \psi(\theta) = \theta^r, r > -n\alpha_0, \alpha_0$ is a known positive constant;
 - (viii) $f_\theta(x) = \alpha_0 \frac{\theta^{\alpha_0}}{x^{\alpha_0+1}} I(x > \theta), \Theta = (0, \infty), \psi(\theta) = \theta^r, r < n\alpha_0, \alpha_0$ is a known positive constant;
 - (ix) $X_1 \sim \text{Bin}(m_0, \theta), \Theta = (0, 1), \psi(\theta) = \theta^r, r$ is a non-zero integer, m_0 is a known positive integer;
 - (x) $X_1 \sim \text{Poisson}(\theta), \Theta = (0, \infty), \psi(\theta) = (P_\theta(X_1 = 2))^2$. Also find the UMVUE of $\psi(\theta) = \cos(\theta)$;
 - (xi) $X_1 \sim \text{Gamma}(\theta, \alpha_0), \Theta = (0, \infty), \psi(\theta) = \theta^r, r > -n\alpha_0, \alpha_0$ is a known positive constant;
 - (xii) $X_1 \sim \text{Gamma}(\sigma_0, \theta), \Theta = (0, \infty), \psi(\theta) = \theta^n, \sigma_0$ is a known positive constant;
 - (xiii) $f_\theta(x) = \frac{\alpha(x)\theta^x}{c(\theta)}, x = 0, 1, 2, \dots, \Theta = (0, \infty), \psi(\theta) = \theta^r, r$ is a positive integer, $\alpha(x) \geq 0, x = 0, 1, 2, \dots, c(\theta)$ is a normalizing factor for the p.m.f.;
 - (xiv) X_1 follows discrete uniform distribution on the set $\{1, 2, \dots, \theta\}, \Theta = \{1, 2, \dots\}, \psi(\theta)$ is an arbitrary real-valued function defined on Θ ;
 - (xv) X_1 follows discrete uniform distribution on the set $\{1, 2, \dots, \theta\}, \Theta = \{1, 2, \dots\} - \{\theta_0\}, \theta_0$ is a fixed positive integer and $\psi(\theta)$ is an arbitrary real-valued function defined on Θ .
2. Let $X_1, \dots, X_n$ be a random sample from a population with mean $\theta \in \mathbb{R}$ and finite variance. Let δ_U be the UMVUE of $\psi(\theta)$ and let δ_0 be any other unbiased estimator of $\psi(\theta)$. Show that $\text{Cov}_\theta(\delta_U, \delta_0) = \text{Var}_\theta(\delta_U)$.

3. (i) In an estimation problem, suppose that UMVUE of an estimand exists. Show that the UMVUE must be unique;
- (ii) Let δ_i be the UMVUE of $\psi_i(\theta)$, $i = 1, \dots, k$, $\theta \in \Theta$. Show that $\delta_P = \sum_{i=1}^k \delta_i$ is the UMVUE of $\psi(\theta) = \sum_{i=1}^k \psi_i(\theta)$.
4. Let $X_1, \dots, X_n$ be a random sample from p.d.f.

$$f_{\theta}(x) = \begin{cases} \frac{\alpha x^{\alpha-1}}{\theta^{\alpha}}, & \text{if } 0 < x < \theta \\ 0, & \text{otherwise} \end{cases}, \quad \theta \in \Theta = (0, \infty),$$

where α is a known positive constant. Let $\psi : (0, \infty) \rightarrow \mathbb{R}$ be a given function.

- (i) Suppose that $\lim_{\theta \rightarrow 0}(\theta^{n\alpha}\psi(\theta)) = c$, for some non-zero constant c . Show that the UMVUE of $\psi(\theta)$ does not exist;
- (ii) Suppose that $\psi(\theta)$ is an u -estimable differentiable function. Find the UMVUE of $\psi(\theta)$. In particular, find the UMVUEs of $\psi_1(\theta) = e^{-\theta}$ and $\psi_2(\theta) = \theta \ln \theta$.
5. Let $X_1, \dots, X_n$ ($n \geq 3$) be a random sample from $N(\mu, \sigma^2)$, where $\underline{\theta} = (\mu, \sigma) \in \mathbb{R} \times (0, \infty) = \Theta$, say.
- (i) Find the UMVUE of $\psi_r(\underline{\theta}) = \mu^r$, $r = 1, 2, 3, 4$;
- (ii) For estimating σ^r ($r < -n + 1$) under the squared error loss function, find the minimum mean squared error (risk function under the SEL function) estimator of the form $\delta_{\alpha}(\underline{X}) = \alpha(\sum_{i=1}^n (X_i - \bar{X})^2)^{\frac{r}{2}}$, $\alpha > 0$. Find the risk functions of the minimum mean squared estimator and the UMVUE.
- (iii) Find the UMVUEs of $\psi_5(\underline{\theta}) = \frac{\mu}{\sigma}$;
- (iv) Find the UMVUE of $\xi_p(\underline{\theta})$, the p -the quantile of X_1 ;
- (v) Find the UMVUE of $\psi_6(\underline{\theta}) = \Phi(\frac{t_0 - \mu}{\sigma})$, where t_0 is a fixed real constant;
- (vi) Find the UMVUE of $\psi_7(\underline{\theta}) = \frac{1}{\sigma}\phi(\frac{t_0 - \mu}{\sigma})$, where t_0 is a fixed real constant.
6. Let $X_1, \dots, X_n$ ($n \geq 4$) be a random sample from $\text{Exp}(\mu, \sigma)$, where $\underline{\theta} = (\mu, \sigma) \in \mathbb{R} \times (0, \infty) = \Theta$, say.
- (i) Find the UMVUE of $\psi_r(\underline{\theta}) = \mu^r$, $r = 1, 2$;
- (ii) For estimating σ^r ($r < -n + 1$) under the squared error loss function, find the minimum mean squared error (risk function under the SEL function) estimator of the form $\delta_{\alpha}(\underline{X}) = \alpha(\sum_{i=1}^n (X_i - X_{(1)}))^r$, $\alpha > 0$. Find the risk functions of the minimum mean squared estimator and the UMVUE.
- (iii) Find the UMVUE of $\xi_p(\underline{\theta})$, the p -the quantile of X_1 ;
- (iv) Find the UMVUE of $\psi_6(\underline{\theta}) = P_{\underline{\theta}}(X_1 \leq t_0)$, where t_0 is a fixed real constant;
7. Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be independent random samples from $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, respectively, where $\underline{\theta} = (\mu_1, \mu_2, \sigma_1, \sigma_2) \in \mathbb{R}^2 \times (0, \infty)^2 = \Theta$, say.
- (i) Find the UMVUE of $\psi_r(\underline{\theta}) = (\mu_1 - \mu_2)^r$, $r = 1, 2, 3$;

- (ii) Find the UMVUE of $\psi_{4,r}(\underline{\theta}) = \frac{\sigma_1^r}{\sigma_2^r}$, ($r > -n + 1$ and $s < n - 1$);
- (iii) Suppose that it is known apriori that $\sigma_1^2 = k\sigma_2^2$, where k is a known positive constant. Find the UMVUEs of $\psi_5(\underline{\theta}) = (\mu_1 - \mu_2)^3$ and $\psi_6(\underline{\theta}) = \sigma_1^3$.
8. Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be independent random samples from $\text{Exp}(\mu_1, \sigma_1)$ and $\text{Exp}(\mu_2, \sigma_2)$, respectively, where $\underline{\theta} = (\mu_1, \mu_2, \sigma_1, \sigma_2) \in \mathbb{R}^2 \times (0, \infty)^2 = \Theta$, say.
- (i) Find the UMVUE of $\psi_r(\underline{\theta}) = (\mu_1 - \mu_2)^r$, $r = 1, 2$;
- (ii) Find the UMVUE of $\psi_{3,r}(\underline{\theta}) = \frac{\sigma_1^r}{\sigma_2^r}$, ($r > -m + 1$ and $s < n - 1$);
- (iii) Suppose that it is known apriori that $\sigma_1 = k\sigma_2$, where k is a known positive constant. Find the UMVUEs of $\psi_4(\underline{\theta}) = (\mu_1 - \mu_2)^2$ and $\psi_5(\underline{\theta}) = \sigma^2$.
9. Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be independent random samples from $\text{Poisson}(\theta_1)$ and $\text{Poisson}(\theta_2)$, respectively, where $\underline{\theta} = (\theta_1, \theta_2) \in (0, \infty)^2$.
- (i) Using alone the sample $X_1, \dots, X_m$, find the UMVUE of $\psi_1(\theta_1) = P_{\theta_1}(X_1 \in O)$, where O is the set of all odd integers;
- (ii) Find the UMVUE of $\psi_2(\theta) = (\theta_1 - \theta_2)^2$.
10. Let $\mathcal{P}$ be the collection of all Lebesgue p.d.f.s (p.d.f.s of all continuous distributions). Let $X_1, \dots, X_n$ be a random sample from an unknown distribution in $\mathcal{P}$.
- (i) Show that $\underline{T} = (X_{(1)}, \dots, X_{(n)})$ is a complete-sufficient statistic;
- (ii) Find the UMVUEs of $\theta = E_f(X_1)$, $\sigma^2 = \text{Var}(X_1)$, $\tau_1 = \theta^2$ and $\tau_2 = P_f(X_1 \leq t_0)$, where t_0 is a fixed real constant.
11. Let $\mathcal{P}$ be the collection of all symmetric Lebesgue p.d.f.s having finite mean. Let $X_1, \dots, X_n$ be a random sample from an unknown distribution in $\mathcal{P}$.
- (i) Is $\underline{T} = (X_{(1)}, \dots, X_{(n)})$ a complete statistic;
- (ii) Show that, for $n \geq 2$, no UMVUE of $\theta = E_f(X_1)$ exists;
- (iii) Show that, for $n = 1$, X_1 is complete for $\mathcal{P}$. In this case, show that $\delta_0(X_1) = X_1$ is the UMVUE of $\tau = E_f(X_1)$.

STAT 418a: Inference-I

Assignment No. 6: Unbiased Estimation, UMVUE, Lehman-Scheffe Theorem

Solutions

Problem No. 1

(i) $\bar{X}$ is C.S. $Z = \frac{\sqrt{n}(\bar{X}-\theta)}{\sigma_0} \sim N(0,1)$

$$E_{\theta}(\bar{X}) = \theta;$$

$$E_{\theta}(\bar{X}^2 - \frac{\sigma_0^2}{n}) = \theta^2$$

$$E_{\theta}((\bar{X}-\theta)^3) = 0 \Rightarrow E_{\theta}(\bar{X}^3) - 3E_{\theta}(\bar{X}^2)\theta + 3E_{\theta}(\bar{X})\theta - \theta^3 = 0$$

$$\Rightarrow E_{\theta}(\bar{X}^3 - \frac{3\sigma_0^2}{n}\bar{X}) = \theta^3$$

$$E_{\theta}((\bar{X}-\theta)^4) = \frac{3\sigma_0^4}{n^2} \Rightarrow E_{\theta}(\bar{X}^4) - 4\theta E_{\theta}(\bar{X}^3) + 6\theta^2 E_{\theta}(\bar{X}^2) - 4\theta^3 E_{\theta}(\bar{X}) + \theta^4 = \frac{3\sigma_0^4}{n^2}$$

$$\Rightarrow E_{\theta}(\bar{X}^4 - \frac{6\sigma_0^2}{n}\bar{X}^2 + \frac{3\sigma_0^4}{n^2}) = \theta^4$$

$\delta_1 = \bar{X}$ is the UMVUE of $\psi_1(\theta) = \theta$

$\delta_2 = \bar{X}^2 - \frac{\sigma_0^2}{n}$ is the UMVUE of $\psi_2(\theta) = \theta^2$

$\delta_3 = \bar{X}^3 - \frac{3\sigma_0^2}{n}\bar{X}$ is the UMVUE of $\psi_3(\theta) = \theta^3$

$\delta_4 = \bar{X}^4 - \frac{6\sigma_0^2}{n}\bar{X}^2 + \frac{3\sigma_0^4}{n^2}$ is the UMVUE of $\psi_4(\theta) = \theta^4$

(ii) $T = \sum_{i=1}^n (X_i - \mu_0)^2$ is C.S. and $T \sim \theta^2 \chi_n^2$

$$E(T^{r/2}) = \theta^r E(\chi_n^2)^{r/2} = \frac{\Gamma(\frac{n-r}{2})}{\Gamma(\frac{n}{2})} 2^{r/2} \theta^r, \quad r > -n$$

$$\Rightarrow \delta_T = \frac{\Gamma(\frac{n}{2})}{2^{r/2} \Gamma(\frac{n-r}{2})} \left[\sum_{i=1}^n (X_i - \mu_0)^2 \right]^{r/2} \text{ is the UMVUE of } \psi_r(\theta) = \theta^r$$

(iii) $T = \sum_{i=1}^n (X_i - \mu_0)$ is C-S and $T \sim \theta \text{Gamma}(n)$

$$E(T^r) = \frac{\theta^r}{\Gamma(n)} \int_0^\infty t^r e^{-t\theta} dt = \frac{\Gamma(n+r)}{\Gamma(n)} \theta^{-r}$$

$\Rightarrow \delta_r = \frac{\Gamma(n)}{\Gamma(n+r)} T^r$ is the UMVUE of θ^{-r} .

(iv) $T = \sum_{i=1}^n (X_i - \mu_0)$ is C-S and $T \sim \theta \text{Gamma}(n)$

$$\begin{aligned} \psi(\theta) &= P_\theta(X_1 - \mu_0 \leq t_0 - \mu_0) \\ &= E_\theta(I(X_1 - \mu_0 \leq t_0 - \mu_0)) \end{aligned}$$

Clearly, for $t_0 < \mu_0$, $\psi(\theta) = 0$.

The UMVUE of $\psi(\theta)$ is (for $t_0 > \mu_0$)

$$\eta(T) = E_\theta(I(X_1 - \mu_0 < t_0 - \mu_0) | T)$$

For $t > 0$,

$$\eta(T) = P(X_1 - \mu_0 < t_0 - \mu_0 \mid \sum_{i=1}^n (X_i - \mu_0) = t)$$

$$= P\left(\frac{X_1 - \mu_0}{\sum_{i=1}^n (X_i - \mu_0)} < \frac{t_0 - \mu_0}{t} \mid \sum_{i=1}^n (X_i - \mu_0) = t\right)$$

$$= P\left(\frac{X_1 - \mu_0}{\sum_{i=1}^n (X_i - \mu_0)} < \frac{t_0 - \mu_0}{t}\right) \quad (\text{Bayes' Theorem})$$

$\hookrightarrow \text{Beta}(1, n-1)$

$$= \begin{cases} 0 & t_0 < \mu_0 \\ \frac{t_0 - \mu_0}{t} & \mu_0 \leq t_0 < \mu_0 + t \\ \int_0^{\frac{t_0 - \mu_0}{t}} (n-1)(1-b)^{n-2} db & t_0 \geq \mu_0 + t \\ 1 & t > t_0 - \mu_0 \end{cases}$$

$$= \begin{cases} 1 - \left(\frac{t_0 - \mu_0}{t}\right)^{n-1} & 0 < t \leq t_0 - \mu_0 \\ 1 & \text{elsewhere} \end{cases}$$

$t_0 < \mu_0$

$\mu_0 \leq t_0 < \mu_0 + t$

$t_0 \geq \mu_0 + t$

$t > t_0 - \mu_0$

$0 < t \leq t_0 - \mu_0$

$$(V) T = x_{(1)} \text{ is C-S. } Z = \frac{n(x_{(1)} - \theta)}{\sigma_0} \sim \text{Exp}(1)$$

$$E(Z) = 1 \Rightarrow E(x_{(1)} - \frac{\sigma_0}{n}) = 0$$

$$E(Z^2) = 2 \Rightarrow E_0[x_{(1)}^2] = \frac{\sigma_0^2}{n^2} + (\theta + \frac{\sigma_0}{n})^2 = \theta^2 + 2\frac{\sigma_0}{n}\theta + \frac{2\sigma_0^2}{n^2}$$

$$\Rightarrow E_0(x_{(1)}^2 - \frac{2\sigma_0}{n}x_{(1)}) = \theta^2$$

$$E(Z^3) = 6 \Rightarrow E[(x_{(1)} - \theta)^3] = \frac{6\sigma_0^3}{n^3}$$

$$\Rightarrow E_0[x_{(1)}^3] - 3\theta(\theta^2 + 2\frac{\sigma_0}{n}\theta + \frac{2\sigma_0^2}{n^2}) + 3\theta^2(\theta + \frac{\sigma_0}{n}) - \theta^3 = \frac{6\sigma_0^3}{n^3}$$

$$\Leftrightarrow E_0[x_{(1)}^3 - \frac{3\sigma_0}{n}(x_{(1)}^2 - \frac{2\sigma_0}{n}x_{(1)}) - \frac{6\sigma_0^2}{n^2}(x_{(1)} - \frac{\sigma_0}{n}) - \frac{6\sigma_0^3}{n^3}] = \theta^3$$

$$\Leftrightarrow E_0[x_{(1)}^3 - \frac{3\sigma_0}{n}x_{(1)}^2] = \theta^3$$

$$E(Z^4) = 24 \Rightarrow E_0[(x_{(1)} - \theta)^4] = \frac{24\sigma_0^4}{n^4}$$

$$\Rightarrow E_0[x_{(1)}^4] - 4\theta(\theta^3 + 3\frac{\sigma_0}{n}\theta^2 + \frac{6\sigma_0^2}{n^2}\theta + \frac{6\sigma_0^3}{n^3})$$

$$+ 6\theta^2(\theta^2 + 2\frac{\sigma_0}{n}\theta + \frac{2\sigma_0^2}{n^2}) - 4\theta^3(\theta + \frac{\sigma_0}{n}) + \theta^4 = \frac{24\sigma_0^4}{n^4}$$

$$\Rightarrow E_0[x_{(1)}^4 - \frac{4\sigma_0}{n}(x_{(1)}^3 - \frac{3\sigma_0}{n}x_{(1)}^2) - \frac{12\sigma_0^2}{n^2}(x_{(1)}^2 - \frac{2\sigma_0}{n}x_{(1)}) - \frac{24\sigma_0^3}{n^3}(x_{(1)} - \frac{\sigma_0}{n})] = \theta^4$$

$$\Rightarrow E_0[x_{(1)}^4 - \frac{4\sigma_0}{n}x_{(1)}^3] = \theta^4 \quad \frac{24\sigma_0^4}{n^4} = \theta^4$$

$$\Rightarrow \delta_1 = x_{(1)} - \frac{\sigma_0}{n} \text{ is the UMVUE of } \theta$$

$$\delta_2 = x_{(1)}^2 - \frac{2\sigma_0}{n}x_{(1)} \text{ is the UMVUE of } \theta^2$$

$$\delta_3 = x_{(1)}^3 - \frac{3\sigma_0}{n}x_{(1)}^2 \text{ is the UMVUE of } \theta^3$$

$$\delta_4 = x_{(1)}^4 - \frac{4\sigma_0}{n}x_{(1)}^3 \text{ is the UMVUE of } \theta^4.$$

(VI) $T = \max_{1 \leq i \leq n} |X_i - \mu_0|$ is the C-S statistic.

$$b_T(t|\theta) = \frac{n t^{n-1}}{\theta^n}, \quad 0 < t < \theta$$

$$E_\theta(T^r) = \frac{n}{n+r} \theta^r, \quad r > -n$$

$\Rightarrow \delta = \frac{n+r}{n} T^r$ is the UMVUE of θ^r , $r > -n$.

(VII) $T = \max\{X_1, \dots, X_n\} = X_{(n)}$ is the C-S statistic.

$$b_T(t|\theta) = \frac{n \theta_0 + \frac{n \theta_0 - 1}{\theta^{n \theta_0}}}{\theta^{n \theta_0}}, \quad 0 < t < \theta$$

$$E_\theta(X_{(n)}^r) = \frac{n \theta_0}{n \theta_0 + r} \theta^r, \quad r > -n \theta_0$$

$\Rightarrow \delta(\delta) = \frac{n \theta_0 + r}{n \theta_0} X_{(n)}^r$ is the UMVUE of $\psi_r(\theta) = \theta^r$

(VIII) $T = X_{(1)}$ is a C-S statistic.

$$b_T(t|\theta) = \frac{n \theta_0 \theta^{n \theta_0}}{t^{n \theta_0 + 1}}, \quad t > \theta$$

$$E_\theta(X_{(1)}^r) = \frac{n \theta_0}{n \theta_0 - r} \theta^r, \quad r < n \theta_0$$

$\Rightarrow \delta(\delta) = \frac{n \theta_0 - r}{n \theta_0} X_{(1)}^r$ is the UMVUE of $\psi_r(\theta) = \theta^r$

(IX) $T = \sum_{i=1}^n X_i$ is a C-S statistic and $T \sim \text{Bin}(n, \theta)$

$$E_\theta(\delta(T)) = \theta^r, \quad \forall \theta \in (0, 1)$$

$$\Leftrightarrow \sum_{k=0}^n \delta(k) \binom{n}{k} \theta^k (1-\theta)^{n-k} = \theta^r, \quad \forall \theta \in (0, 1)$$

$$\begin{aligned} \Leftrightarrow \sum_{k=0}^n \delta(k) \binom{n}{k} p^k &= (1+p)^n \left(\frac{p}{1+p}\right)^r, \quad \forall p \in (0, \infty) \dots (A) \\ &= p^r \sum_{j=0}^{n-r} \binom{n-r}{j} p^j, \quad \forall p \in (0, \infty), \quad r \leq n \end{aligned}$$

$$\Leftrightarrow \sum_{k=0}^n \delta(k) \binom{n}{k} p^k = \sum_{j=r}^n \binom{n-r}{j-r} p^j, \quad \forall p \in (0, \infty) \dots (*)$$

4/30

Case I $r \leq -1$

In this case (A) will never be ~~unbiased~~ ^{Admitted}

$\Rightarrow$ Unbiased estimator of $\psi_r(\theta) = \theta^r$ based on C-S Statistic

$T = \sum_{i=1}^n x_i$ does not exist

$\Rightarrow$ UMVUE of $\psi_r(\theta) = \theta^r$ (in fact any unbiased estimator of $\psi_r(\theta) = \theta^r$) does not exist

Case II $1 \leq r \leq m+n$

From (A), we get

$$g(k) = \begin{cases} 0, & k = 0, 1, 2, \dots, r-1 \\ \frac{\binom{m+n-r}{k-r}}{\binom{m+n}{k}}, & k = r, r+1, \dots, m+n \end{cases}$$

and $g(T)$ is the UMVUE of $\psi_r(\theta) = \theta^r$

Case III $r \geq m+n+1$

From (A), we get

$$\begin{aligned} \sum_{k=0}^{m+n} g(k) \binom{m+n}{k} p^k &= p^r (1+p)^{-(r-m-n)}, \quad \forall p \in (0, 1) \\ &= p^r \sum_{j=0}^{\infty} \binom{r-m-n}{j} (-1)^j p^j, \quad \forall p \in (0, 1) \\ &= \sum_{j=r}^{\infty} \binom{r-m-n}{j-r} (-1)^{j-r} p^j, \quad \forall p \in (0, 1) \end{aligned}$$

$\rightarrow$ Can not be Admitted

$\Rightarrow$ Unbiased estimator of θ^r based on C-S Statistic $T = \sum_{i=1}^n x_i$ does not exist

$\Rightarrow$ Any unbiased estimator of θ^r does not exist

$\Rightarrow$ UMVUE of θ^r does not exist.

(X) $T = \sum_{i=1}^n X_i$ is c.s and $T \sim \text{Poisson}(\theta)$

$$\psi(\theta) = (P_\theta(X_1=2))^2 = \left[\frac{e^{-\theta} \theta^2}{2} \right]^2 = \frac{e^{-2\theta} \theta^4}{4}$$

The UMVUE of $\psi(\theta)$ is $g(T)$, where

$$E_\theta(g(T)) = \psi(\theta), \quad \forall \theta > 0$$

$$\Rightarrow \sum_{k=0}^{\infty} g(k) \frac{e^{-\theta} \theta^k}{k!} = \frac{e^{-2\theta} \theta^4}{4}, \quad \forall \theta > 0$$

$$\begin{aligned} \Rightarrow \sum_{k=0}^{\infty} \frac{g(k) \theta^k}{k!} &= \frac{\theta^4}{4} e^{-\theta} \\ &= \frac{\theta^4}{4} \sum_{j=0}^{\infty} \frac{(-1)^j \theta^j}{j!} \\ &= \sum_{j=4}^{\infty} \frac{(-1)^j \theta^j}{4 j!} \quad , \quad \forall \theta > 0 \end{aligned}$$

$$\Rightarrow g(k) = \begin{cases} 0, & k=0, 1, 2, 3 \\ \frac{(-1)^k}{4 k!}, & k=4, 5, 6, \dots \end{cases}$$

The UMVUE of $C_{\theta 1}$ is $g_1(T)$, where

$$E_\theta(g_1(T)) = C_{\theta 1}, \quad \forall \theta > 0$$

$$\Rightarrow \sum_{k=0}^{\infty} g_1(k) \frac{e^{-\theta} \theta^k}{k!} = C_{\theta 1}, \quad \forall \theta > 0$$

$$\Rightarrow \sum_{k=0}^{\infty} \frac{g_1(k) \theta^k}{k!} = e^{\theta} C_{\theta 1}, \quad \forall \theta > 0$$

$$= \left(\sum_{j=0}^{\infty} \frac{\theta^j}{j!} \right) \left(\sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k}}{2k!} \right), \quad \forall \theta > 0$$

$$= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{j+2k}}{j! 2k!} \quad \left(\begin{array}{l} j=m \\ j+2k=l; \quad k=\frac{l-m}{2} \\ j=m \end{array} \right)$$

$$= \sum_{l=0}^{\infty} \left\{ \sum_{m \in \{0, 2, 4, \dots, l\}} \frac{(-1)^{\frac{l-m}{2}}}{m! l!} \right\} \theta^l, \quad \forall \theta > 0$$

where $a_l = \begin{cases} 0, & \text{if } l \text{ is even} \\ 1, & \text{if } l \text{ is odd} \end{cases}$

$$\Rightarrow \frac{g(k)}{L^k} = \sum_{m \in \{a_1, a_2, \dots, k\}} \frac{(-1)^{\frac{k-m}{2}}}{L^m L^{k-m}}, \quad k=0, 1, 2, \dots$$

$$\Rightarrow g(k) = \sum_{m \in \{a_1, a_2, \dots, k\}} \binom{k}{m} (-1)^{\frac{k-m}{2}}, \quad k=0, 1, 2, \dots$$

(X) $T = \sum_{i=1}^n x_i$ is a C-S Statistic and $T \sim \theta \text{ Gamma}(1, n\alpha_0)$

$$E_\theta(T^r) = \frac{\Gamma(n\alpha_0 r)}{\Gamma(n\alpha_0)} \theta^r, \quad r > -n\alpha_0$$

$$\Rightarrow g = \frac{\Gamma(n\alpha_0)}{\Gamma(n\alpha_0 r)} T^r \text{ is the UMVUE of } \psi_r(\theta) = \theta^r$$

(XII) The joint p.d.f. of $\underline{x} = (x_1, \dots, x_n)$ is

$$g_\theta(\underline{x}) = \frac{1}{\sigma_0^{n\theta} (\Gamma\theta)^n} \frac{e^{-\sum_{i=1}^n \frac{x_i}{\sigma_0}}}{(\prod_{i=1}^n x_i)} e^{\theta \ln(\prod_{i=1}^n x_i)} \prod_{i=1}^n I(x_i > 0), \quad \theta > 0$$

$\Rightarrow T(\underline{x}) = \prod_{i=1}^n x_i$ is a C-S Statistic.

$$E_\theta(T(\underline{x})) = \prod_{i=1}^n E_\theta(x_i) = \sigma_0^n \theta^n$$

$$\Rightarrow g = \prod_{i=1}^n \frac{x_i}{\sigma_0} \text{ is the UMVUE of } \psi(\theta) = \theta^n.$$

(XIII) The joint p.d.f. of $\underline{x} = (x_1, \dots, x_n)$ is

$$g_\theta(\underline{x}) = \frac{\prod_{i=1}^n \alpha(x_i)}{[C(\theta)]^n} e^{(\eta\theta) \sum_{i=1}^n x_i} \prod_{i=1}^n I(x_i \in \{0, 1, 2, \dots, \eta\}), \quad \theta > 0$$

$\eta = \lfloor \theta \rfloor \in (-\infty, \infty)$ contains an open rectangle in it.

$\Rightarrow T(\underline{x}) = \sum_{i=1}^n x_i$ is a C-S Statistic.

The p.m.f of $T = \sum_{i=1}^n X_i$ is

$$\begin{aligned}
 b_T(t|\theta) &= P_\theta(T(X) = t), \quad t = 0, 1, 2, \dots \\
 &= \sum_{\lambda_1=0}^{\infty} \dots \sum_{\lambda_n=0}^{\infty} \frac{\left(\prod_{i=1}^n \alpha(\lambda_i) \right)}{(c(\theta))^n} \theta^{\sum \lambda_i} \\
 &= \frac{\theta^t}{(c(\theta))^n} \sum_{\lambda_1=0}^{\infty} \dots \sum_{\lambda_n=0}^{\infty} \left(\prod_{i=1}^n \alpha(\lambda_i) \right) \\
 &\quad \sum_{i=1}^n \lambda_i = t \\
 &= \frac{A_n(t) \theta^t}{(c(\theta))^n}, \quad t = 0, 1, 2, \dots
 \end{aligned}$$

Where $A_n(t) = \sum_{\lambda_1=0}^{\infty} \dots \sum_{\lambda_n=0}^{\infty} \left(\prod_{i=1}^n \alpha(\lambda_i) \right)$
 $\sum_{i=1}^n \lambda_i = t$

Or $A_n(t) =$ Coeff. of θ^t in power series expansion of $(c(\theta))^n$ (Since $\sum_{t=0}^{\infty} A_n(t) \theta^t = (c(\theta))^n$).

The UMVUE of $\psi_r(\theta) = \theta^r$ is $\delta_r(T)$ where

$$E_\theta(\delta_r(T)) = \theta^r, \quad \forall \theta > 0$$

$$\Leftrightarrow \frac{1}{(c(\theta))^n} \sum_{k=0}^{\infty} \delta_r(k) A_n(k) \theta^k = \theta^r, \quad \forall \theta > 0$$

$$\Leftrightarrow \sum_{k=0}^{\infty} \delta_r(k) A_n(k) \theta^k = \theta^r (c(\theta))^n = \theta^r \sum_{j=0}^{\infty} A_n(j) \theta^j, \quad \forall \theta > 0$$

$$\Leftrightarrow \sum_{k=0}^{\infty} \delta_r(k) A_n(k) \theta^k = \sum_{k=r}^{\infty} A_n(k-r) \theta^k, \quad \forall \theta > 0$$

$$\Leftrightarrow \delta_r(k) = \begin{cases} 0, & k = 0, 1, \dots, r-1 \\ \frac{A_n(k-r)}{A_n(k)}, & k = r, r+1, \dots \end{cases}$$

(XIV) $T = x_{(n)}$ is C-S for $\theta \in \Theta = \{1, 2, \dots\}$

$$P_\theta(T \leq k) = P_\theta(x_{(n)} \leq k) = \left(\frac{k}{\theta}\right)^n, \quad k \in \{1, 2, \dots, \theta\}$$

$$P_\theta(T = k) = P_\theta(x_{(n)} \leq k) - P_\theta(x_{(n)} \leq k-1) \\ = \left(\frac{k}{\theta}\right)^n - \left(\frac{k-1}{\theta}\right)^n, \quad k = 1, 2, \dots, \theta, \quad \theta \in \{1, 2, \dots\}$$

The UMVUE of $\psi(\theta)$ is $g(T)$, where

$$E_\theta(g(T)) = \psi(\theta), \quad \forall \theta \in \{1, 2, \dots\}$$

$$\Rightarrow \sum_{k=1}^{\theta} g(k) [k^n - (k-1)^n] = \theta^n \psi(\theta), \quad \theta \in \{1, 2, \dots\}$$

$$\& \sum_{k=1}^{\theta+1} g(k) [k^n - (k-1)^n] = (\theta+1)^n \psi(\theta+1), \quad \theta \in \{1, 2, \dots\}$$

$$\Rightarrow g(1) = \psi(1) \quad \text{and} \quad g(\theta+1) [(\theta+1)^n - \theta^n] = (\theta+1)^n \psi(\theta+1) - \theta^n \psi(\theta), \quad \theta \in \{1, 2, \dots\}$$

$$\Rightarrow g(k) = \frac{k^n \psi(k) - (k-1)^n \psi(k-1)}{k^n - (k-1)^n}, \quad k = 2, 3, \dots \quad g(1) = \psi(1)$$

(XV) Note that under $\theta \in \Theta_0 = \{1, 2, \dots\} - \{\theta_0\}$, $T = x_{(n)}$ is not-complete.

It suffices to consider estimators based on minimal sufficient statistic $T = x_{(n)}$. Here $U \in \mathcal{U}$ (the class of unbiased estimators of zero) i.e.

$$E_\theta(U(x_{(n)})) = 0, \quad \forall \theta \in \{1, 2, \dots, \theta_0-1, \theta_0+1, \dots\}$$

$$\Rightarrow \sum_{k=1}^{\theta} U(k) [k^n - (k-1)^n] = 0, \quad \forall \theta \in \{1, 2, \dots, \theta_0-1, \theta_0+1, \dots\}$$

$$\Rightarrow U(k) = 0, \quad k = 1, \dots, \theta_0-1, \theta_0+1, \dots$$

$$\text{and } U(\theta_0) (\theta_0^n - (\theta_0-1)^n) + U(\theta_0+1) ((\theta_0+1)^n - \theta_0^n) = 0$$

$$\Rightarrow U(k) = \begin{cases} 0, & k = 1, \dots, \theta_0-1, \theta_0+1, \dots \\ c, & k = \theta_0 \\ -c \frac{\theta_0^n - (\theta_0-1)^n}{(\theta_0+1)^n - \theta_0^n}, & k = \theta_0+1 \end{cases} \dots (*)$$

An estimator g will be ^{UMVUE} unbiased for $\psi(\theta) = E(g(x_1))$ if

$$E(g(x_1)u(x_1)) = 0, \quad \forall u = 1, \dots, \theta_0, \theta_0+1, \dots$$

& u of the type (*)

$$g(k)u(k) = 0, \quad \forall k = 1, 2, \dots, \theta_0, \theta_0+1, \dots$$

$$g(\theta_0)u(\theta_0) = d$$

$$g(\theta_0+1)u(\theta_0+1) = -d \frac{\theta_0^2 - (\theta_0+1)^2}{(\theta_0+1)^2 - \theta_0^2}$$

for every u of the type (*)

$$\Rightarrow g(\theta_0) = r, \quad g(\theta_0+1) = -r, \quad g(k) = \text{arbitrary}$$

Problem No. 2

$U = g_U - g_0 \in \mathcal{U}$ (the class of unbiased estimators of zero)

$$g_U \text{ is UMVUE} \Rightarrow \text{Cov}_\theta(g_U, U) = 0, \quad \forall \theta \in \Theta$$

$$\Rightarrow \text{Var}_\theta(g_U) = \text{Cov}_\theta(g_U, g_0).$$

Problem No. 3

(i) Let g_1 and g_2 be two UMVUEs of $\psi(\theta)$. Then $E(g_i) = \psi(\theta), \forall \theta \in \Theta$ and $\text{Var}_\theta(g_1) = \text{Var}_\theta(g_2), \forall \theta \in \Theta = V(\theta), \forall \theta$

Consider the estimator $g = \frac{g_1 + g_2}{2}$. Then g is unbiased for $\psi(\theta)$ and

$$\text{Var}_\theta(g) = \frac{1}{4} [\text{Var}_\theta(g_1) + \text{Var}_\theta(g_2) + 2\text{Cov}_\theta(g_1, g_2)]$$

$$\leq \frac{1}{4} [V(\theta) + V(\theta) + 2\sqrt{V(\theta)V(\theta)}] = V(\theta) = \text{Var}_\theta(g_i), \quad \forall \theta \in \Theta$$

and equality is attained iff

$$g_1 - \psi(\theta) = k(\theta)(g_2 - \psi(\theta)) \quad \text{w.p.1}$$

$$\text{Var}_\theta(g_1) = \text{Var}_\theta(g_2), \quad \forall \theta \in \Theta$$

$$\Rightarrow |k(\theta)| = 1 \Rightarrow k(\theta) = \pm 1, \quad \forall \theta \in \Theta$$

An g_1 and g_2 are statistics $k(\theta) = -1$ is ruled out (as it leads to $g_1 + g_2 = 2\psi(\theta)$)

10/30

Thus we have equality iff $\delta_1 = \delta_2$ w.p.1 or, in other words,
 if $P_\theta(\delta_1 = \delta_2) \neq 0, \forall \theta \in \Theta$, then

$$\text{Var}_\theta(\delta) < \text{Var}_\theta(\delta_i), \quad i=1, 2,$$

Contradicting that δ_i are the UMVUE's.

(ii) δ_i is the UMVUE of $\psi_i(\theta)$ implies that, for $i=1, \dots, k$
 $E_\theta(\delta_i) = \psi_i(\theta), \forall \theta \in \Theta$ and $\text{Cov}_\theta(\delta_i, u) = 0, \forall u \in \mathcal{U}$.

Clearly

$$E_\theta(\delta_p) = \sum_{i=1}^k E_\theta(\delta_i) = \sum_{i=1}^k \psi_i(\theta) = \psi(\theta), \quad \forall \theta \in \Theta$$

And, for any $u \in \mathcal{U}$

$$\text{Cov}_\theta(\delta_p, u) = \text{Cov}_\theta\left(\sum_{i=1}^k \delta_i, u\right) = \sum_{i=1}^k \text{Cov}_\theta(\delta_i, u) = 0, \quad \forall \theta \in \Theta$$

$\Rightarrow \delta_p$ is the UMVUE of $E_\theta(\delta_p) = \psi(\theta)$.

Problem No. 4

$T = X_{(n)}$ is C-S and the p.d.f. of T is
 $f_T(t|\theta) = \frac{n\alpha t^{n\alpha-1}}{\theta^{n\alpha}}, \quad 0 < t < \theta$.

(i) We know that if the UMVUE exists then it must be a function of C-S statistic $T = X_{(n)}$, i.e. we must have $\delta(T)$ such that

$$E_\theta(\delta(T)) = \psi(\theta), \quad \forall \theta \in \Theta$$

$$\Leftrightarrow \frac{n\alpha}{\theta^{n\alpha}} \int_0^\theta \delta(t) t^{n\alpha-1} dt = \psi(\theta), \quad \forall \theta > 0$$

$$\Rightarrow n\alpha \int_0^\theta \delta(t) t^{n\alpha-1} dt = \theta^{n\alpha} \psi(\theta), \quad \forall \theta > 0$$

$$\Rightarrow \lim_{\theta \rightarrow 0} \left(n\alpha \int_0^\theta \delta(t) t^{n\alpha-1} dt \right) = C$$

$$\Rightarrow C = 0 \quad (\text{a contradiction})$$

Thus the UMVUE does not exist.

(iii) The UMVUE of α -estimable function $\psi(\alpha)$ is $g(\tau)$, where

$$\begin{aligned} E_{\theta}(g(\tau)) &= \psi(\alpha), \quad \forall \theta > 0 \\ \Rightarrow \alpha \int_0^{\infty} g(t) t^{\alpha-1} dt &= \theta^{\alpha} \psi(\alpha), \quad \forall \theta > 0 \\ \Rightarrow \alpha g(0) \theta^{\alpha-1} &= \theta^{\alpha} \psi'(\alpha) + \alpha \theta^{\alpha-1} \psi(\alpha), \quad \forall \theta > 0 \\ \Rightarrow g(0) &= \frac{1}{\alpha} [\theta \psi'(\alpha) + \alpha \psi(\alpha)] \\ &= \frac{\theta \psi'(\alpha)}{\alpha} + \psi(\alpha) \end{aligned}$$

i.e. $g(\tau) = \frac{\tau \psi'(\tau)}{\alpha} + \psi(\tau)$

Obviously, $E_{\theta}(g(\tau)) = \psi(\alpha), \quad \forall \theta > 0$.

The UMVUE of $\psi_1(\alpha) = e^{-\alpha}$ is

$$g_1(\tau) = \left[e^{-\tau} - \frac{\tau e^{-\tau}}{\alpha} \right] = \left[1 - \frac{\tau}{\alpha} \right] e^{-\tau}$$

The UMVUE of $\psi_2(\alpha) = \alpha \ln \alpha$ is

$$g_2(\tau) = \tau \ln \tau + \frac{\tau}{\alpha} [\ln \tau + 1] = \frac{\tau}{\alpha} + \frac{\alpha-1}{\alpha} \tau \ln \tau$$

Problem No. 5

$T = (T_1, T_2) = (\bar{X}, \sum_{i=1}^n (X_i - \bar{X})^2)$ is a CS statistic

$$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$$

$$\sum_{i=1}^n (X_i - \bar{X})^2 \sim \sigma^2 \chi_{n-1}^2 \quad \left. \begin{array}{l} \bar{X} \\ \sum_{i=1}^n (X_i - \bar{X})^2 \end{array} \right\} \text{independent}$$

(i) By Problem (i)

$$E_{\theta}(\bar{X}) = \mu$$

$$E_{\theta}(\bar{X}^2 - \frac{\sigma^2}{n}) = \mu^2$$

$$E(\bar{x}^3 - \frac{3\sigma^2}{n} \bar{x}) = \mu^3$$

$$E(\bar{x}^4 - \frac{6\sigma^2}{n^2} \bar{x}^2 + \frac{3\sigma^4}{n^2}) = \mu^4$$

Also $\bar{x}$ and $S_{n-1}^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ are independent with

$$E_{\theta}(S_{n-1}^2) = \sigma^2 \quad \text{and} \quad E_{\theta}((n-1)^{-1} S_{n-1}^4) = \sigma^4 E((X_{n-1}^L)^2) \\ = \sigma^4 [2(n-1) + (n-1)^2] \\ = (n^2 - 1)\sigma^4$$

$$\text{i.e. } E_{\theta}(S_{n-1}^2) = \sigma^2 \quad \text{and} \quad E_{\theta}(\frac{n-1}{n^2} S_{n-1}^4) = \sigma^4$$

$\Rightarrow$ The UMVUE of $\psi_1(\theta) = \mu$ is $\delta_1(X) = \bar{x}$

The UMVUE of $\psi_2(\theta) = \mu^2$ is $\delta_2(X) = \bar{x}^2 - \frac{S_{n-1}^2}{n}$

The UMVUE of $\psi_3(\theta) = \mu^3$ is $\delta_3(X) = \bar{x}^3 - \frac{3S_{n-1}^2}{n} \bar{x}$

The UMVUE of $\psi_4(\theta) = \mu^4$ is $\delta_4(X) = \bar{x}^4 - \frac{6S_{n-1}^2}{n^2} \bar{x}^2 + \frac{3(n-1)}{n^2(n-1)} S_{n-1}^4$

(iii) For any $\theta \in \Theta$

$$R(\theta, \delta) = E_{\theta} \left[\left(\alpha \left(\sum_{i=1}^n (x_i - \bar{x})^2 \right)^{r/2} - \sigma^r \right)^2 \right]$$

is minimized at

$$\alpha \equiv \alpha(\theta) = \frac{\sigma^r E_{\theta} \left[\left(\sum_{i=1}^n (x_i - \bar{x})^2 \right)^{r/2} \right]}{E_{\theta} \left[\left(\sum_{i=1}^n (x_i - \bar{x})^2 \right)^r \right]}$$

$$= \frac{\sigma^r E \left[\left(\sum_{i=1}^n x_{n-1}^L \right)^{r/2} \right]}{E \left[\left(\sum_{i=1}^n x_{n-1}^L \right)^r \right]}$$

$$= \frac{E \left[(X_{n-1}^L)^{r/2} \right]}{E \left[(X_{n-1}^L)^r \right]} = \frac{\Gamma \frac{n-1}{2} + \frac{r}{2}}{2^{r/2} \Gamma \frac{n-1}{2} + r} = C_{r,n}, \quad \wedge$$

The minimum mean squared estimator is

$$\delta_M(X) = C_{r,n} \left(\sum_{i=1}^n (x_i - \bar{x})^2 \right)^{r/2}$$

$$\left(\sum_{i=1}^n (x_i - \bar{x})^2 \right)^{1/2} \sim \sigma^2 (X_{n-1}^2)^{1/2}$$

$$E \left[\left(\sum_{i=1}^n (x_i - \bar{x})^2 \right)^{1/2} \right] = \frac{\sigma^2 \frac{2^{1/2} \sqrt{\frac{n-1}{2} + \frac{v}{2}}}{\sqrt{\frac{n-1}{2}}}}{\frac{\sigma^2}{dv, n}} = \frac{\sigma^2}{dv, n}$$

The UMVUE of σ^2 is

$$S_U(X) = \frac{\sqrt{\frac{n-1}{2}}}{2^{1/2} \sqrt{\frac{n-1}{2} + \frac{v}{2}}} \left(\sum_{i=1}^n (x_i - \bar{x})^2 \right)^{1/2} = dv, n \left(\sum_{i=1}^n (x_i - \bar{x})^2 \right)^{1/2}$$

$$R(\underline{\theta}, \delta_\alpha) = \alpha^2 E_\theta(T^2) - 2\alpha \sigma^v E_\theta(T) + \sigma^{2v}$$

$$\text{where } T = \left(\sum_{i=1}^n (x_i - \bar{x})^2 \right)^{1/2} \sim \sigma^2 (X_{n-1}^2)^{1/2}$$

$$E_\theta(T) = \frac{2^{1/2} \sqrt{\frac{n-1}{2} + \frac{v}{2}}}{\sqrt{\frac{n-1}{2}}} \sigma^v = \frac{\sigma^v}{dv, n}$$

$$E_\theta(T^2) = \sigma^{2v} E_\theta((X_{n-1}^2)^v) = 2^v \frac{\sqrt{\frac{n-1}{2} + v}}{\sqrt{\frac{n-1}{2}}} \sigma^{2v}$$

$$R(\underline{\theta}, \delta_\alpha) = \alpha^2 \frac{2^v \sqrt{\frac{n-1}{2} + v}}{\sqrt{\frac{n-1}{2}}} \sigma^{2v} - 2\alpha \frac{\sigma^{2v}}{dv, n} + \sigma^{2v}$$

$$= \left(\frac{\alpha^2}{C_{v,n} dv, n} - \frac{2\alpha}{dv, n} + 1 \right) \sigma^{2v}$$

$$R(\underline{\theta}, \delta_\alpha) = \left(1 - \frac{C_{v,n}}{dv, n} \right) \sigma^{2v}$$

$$R(\underline{\theta}, \delta_U) = \left(\frac{dv, n}{C_{v,n}} - 1 \right) \sigma^{2v}$$

(iii) we have $E_\theta(\bar{x}) = \mu$. Also $T = \left(\sum_{i=1}^n (x_i - \bar{x})^2 \right)^{1/2} \sim \sigma^2 X_{n-1}^2$
 $\Rightarrow E_\theta \left(\frac{1}{\sqrt{T}} \right) = \frac{1}{\sigma} E_\theta \left(\frac{1}{\sqrt{X_{n-1}^2}} \right) = \frac{1}{\sigma \sqrt{2}} \frac{\sqrt{\frac{n-1}{2}}}{\sqrt{\frac{n-1}{2}}} = \frac{C_n}{\sigma}, \text{ say,}$

$$\text{where } C_n = \frac{\sqrt{\frac{n-1}{2}}}{\sqrt{2} \sqrt{\frac{n-1}{2}}}$$

[14/30]

Since $\bar{X}$ and $\sum_{i=1}^n (x_i - \bar{X})^2$ are independent, we have

$$E_0\left(\frac{\bar{X}}{\sqrt{T}}\right) = E_0\left(\bar{X} \times \frac{1}{\sqrt{T}}\right) = E_0(\bar{X}) \times E_0\left(\frac{1}{\sqrt{T}}\right) = c_n \frac{\mu}{\sigma}$$

$$\Rightarrow g_5(x) = \frac{1}{c_n} \frac{\bar{X}}{\sqrt{T}} \text{ is the UMVUE of } \psi_5(\theta) = \frac{\mu}{\sigma}.$$

$$= \sqrt{\frac{2}{n-1}} \frac{\sqrt{\frac{n-1}{2}}}{\sqrt{\frac{n-1}{2}}} \frac{\bar{X}}{S_{n-1}}$$

$$\mu + \Phi^{-1}(p)\sigma$$

(iv) $\psi_6(\theta) =$

$$E_0\left(\sqrt{\sum_{i=1}^n (x_i - \bar{X})^2}\right) = \sigma E_0\left(\sqrt{\chi_{n-1}^2}\right) = \sqrt{2} \frac{\sqrt{\frac{n-1}{2}}}{\sqrt{\frac{n-1}{2}}} \sigma$$

$$\Rightarrow E_0\left(\bar{X} + \Phi^{-1}(p) \frac{\sigma}{\sqrt{n}}\right) = \mu + \frac{\sigma}{\sqrt{n}} \Phi^{-1}(p),$$

where $d_n = \frac{\sqrt{n-1} \sqrt{\frac{n-1}{2}}}{\sqrt{2} \sqrt{\frac{n-1}{2}}}$

$$\Rightarrow g_6(x) = \bar{X} + \sqrt{\frac{n-1}{2}} \frac{\sqrt{\frac{n-1}{2}}}{\sqrt{\frac{n-1}{2}}} \Phi^{-1}(p) S_{n-1} \text{ is the UMVUE of}$$

$$\psi_6(\theta) = \mu + \Phi^{-1}(p) \sigma \text{ (the } p\text{-th quantile of } x_1)$$

(v) $\psi_6(\theta) = \Phi\left(\frac{t_0 - \mu}{\sigma}\right) = P_0(X_1 \leq t_0) = E_0(I(X_1 \leq t_0))$

The UMVUE of $\psi_6(\theta)$ is

$$g_6(x) = E_0(I(X_1 \leq t_0) | \bar{X}, \sum_{i=1}^n (x_i - \bar{X})^2)$$

$$E_0[I(X_1 \leq t_0) | \bar{X} = \lambda, \sum_{i=1}^n (x_i - \bar{X})^2 = t]$$

$$= P_0(X_1 \leq t_0 | \bar{X} = \lambda, \sum_{i=1}^n (x_i - \bar{X})^2 = t)$$

$$= P_0\left(\frac{X_1 - \bar{X}}{\sqrt{\sum_{i=1}^n (x_i - \bar{X})^2}} \leq \frac{t_0 - \lambda}{\sqrt{t}} \mid \bar{X} = \lambda, \sum_{i=1}^n (x_i - \bar{X})^2 = t\right)$$

$$= P_0\left(\frac{X_1 - \bar{X}}{\sqrt{\sum_{i=1}^n (x_i - \bar{X})^2}} \leq \frac{t_0 - \lambda}{\sqrt{t}}\right) \quad (\text{Basu's Theorem})$$

$$= F_2\left(\frac{t_0 - \lambda}{\sqrt{t}}\right),$$

Where $F_2(\cdot)$ is the d.f. of r.v. $Z = \frac{X_1 - \bar{X}}{\sqrt{\sum (X_i - \bar{X})^2}}$

It can be shown (seems to be a non-trivial exercise) that the p.d.f of $Z = \frac{X_1 - \bar{X}}{\sqrt{\sum (X_i - \bar{X})^2}}$ is

$$f_Z(z) = \frac{\sqrt{n} \sqrt{\frac{n-1}{2}}}{\sqrt{\pi} (n-1) \sqrt{\frac{n-2}{2}}} \left(1 - \frac{n z^2}{(n-1)^2}\right)^{\frac{n-2}{2}}, \quad |z| \leq \frac{n-1}{\sqrt{n}}.$$

$\Rightarrow S_6(Y) = F_2\left(\frac{t_0 - \bar{X}}{\sqrt{\sum (X_i - \bar{X})^2}}\right)$ is the UMVUE of $\psi_6(\theta) = \Phi\left(\frac{t_0 - \mu}{\sigma}\right)$

Where

$$F_2(z) = \frac{\sqrt{n} \sqrt{\frac{n-1}{2}}}{\sqrt{\pi} (n-1) \sqrt{\frac{n-2}{2}}} \int_{-\infty}^z \left(1 - \frac{n t^2}{(n-1)^2}\right)^{\frac{n-2}{2}} dt.$$

(vi) We have

$$E_\theta(S_6(Y)) = \Phi\left(\frac{t_0 - \mu}{\sigma}\right), \quad \forall t_0, \mu \in \mathbb{R}, \sigma > 0$$

$$\Rightarrow \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F_2\left(\frac{t_0 - \lambda}{\sqrt{t}}\right) b_{\bar{X}}, \tilde{\Sigma}(X_i - \bar{X})^2 \lambda^{n-1} d\lambda dt = \Phi\left(\frac{t_0 - \mu}{\sigma}\right), \quad \forall t_0, \mu, \sigma$$

$$\Rightarrow \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{\sqrt{t}} b_Z\left(\frac{t_0 - \lambda}{\sqrt{t}}\right) b_{\bar{X}}, \tilde{\Sigma}(X_i - \bar{X})^2 \lambda^{n-1} d\lambda dt = \frac{1}{\sigma} \phi\left(\frac{t_0 - \mu}{\sigma}\right), \quad \forall t_0, \mu, \sigma$$

$$\Rightarrow E_\theta \left[\frac{1}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2}} b_Z\left(\frac{t_0 - \bar{X}}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2}}\right) \right] = \frac{1}{\sigma} \phi\left(\frac{t_0 - \mu}{\sigma}\right), \quad \forall t_0, \mu, \sigma$$

$$\Rightarrow S_7(Y) = \frac{1}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2}} b_Z\left(\frac{t_0 - \bar{X}}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2}}\right) \text{ is the UMVUE of}$$

$$\psi_7(\theta) = \frac{1}{\sigma} \phi\left(\frac{t_0 - \mu}{\sigma}\right).$$

Problem No. 6 $(x_{(1)}, \sum_{i=1}^n (x_i - x_{(1)}))$ is C-S,

$$Z_1 = n(x_{(1)} - \mu) \sim \sigma \text{Exp}(1)$$

$$Z_2 = \sum_{i=1}^n (x_i - x_{(1)}) \sim \sigma \text{Gamma}(n-1)$$

$\left. \begin{array}{l} Z_1 \\ Z_2 \end{array} \right\} \text{ independent}$

$$(i) \quad E_{\theta}(Z_1) = \sigma \Rightarrow E_{\theta}(x_{(1)}) = \mu + \frac{\sigma}{n}$$

$$E(Z_2) = (n-1)\sigma$$

$$\Rightarrow E_{\theta} \left[x_{(1)} - \frac{1}{n(n-1)} \sum_{i=1}^n (x_i - x_{(1)}) \right] = \mu, \quad \forall \theta \in \Theta$$

$$\Rightarrow \delta_1(x) = x_{(1)} - \frac{1}{n(n-1)} \sum_{i=1}^n (x_i - x_{(1)}) = x_{(1)} - \frac{S_{n1}}{n} \text{ is the UMVUE of } \mu,$$

$$\text{Where } S_{n1} = \frac{1}{n-1} \sum_{i=1}^n (x_i - x_{(1)})$$

$$\text{Var}(Z_1) = \sigma^2 \Rightarrow \text{Var}_{\theta}(x_{(1)}) = \frac{\sigma^2}{n^2}$$

$$\Rightarrow E_{\theta}(x_{(1)}^2) = \frac{\sigma^2}{n^2} + \left(\mu + \frac{\sigma}{n} \right)^2 = \mu^2 + \frac{2\mu\sigma}{n} + \frac{2\sigma^2}{n^2}$$

$$E(Z_2^2) = n(n-1)\sigma^2$$

$$\Rightarrow E_{\theta}(x_{(1)}^2) = \mu^2 + \frac{2\sigma}{n} E_{\theta}(x_{(1)} - \frac{\sigma}{n}) + \frac{2\sigma^2}{n^2}$$

$$= \mu^2 + \frac{2\sigma}{n} E_{\theta}(x_{(1)})$$

$$= \mu^2 + \frac{2}{n(n-1)} E_{\theta}(Z_2) E_{\theta}(x_{(1)})$$

$$= \mu^2 + \frac{2}{n(n-1)} E_{\theta}(x_{(1)} Z_2)$$

$$\Rightarrow E_{\theta} \left(x_{(1)}^2 - \frac{2}{n(n-1)} x_{(1)} \sum_{i=1}^n (x_i - x_{(1)}) \right) = \mu^2$$

$$\Rightarrow \delta_2(x) = x_{(1)}^2 - \frac{2 S_{n1} x_{(1)}}{n} \text{ is the UMVUE of } \mu^2$$

$$(ii) \quad \delta_{\alpha} = \alpha Z_2^r, \quad \alpha > 0$$

for any fixed $\theta \in \Theta$,

$$R(\theta, \delta_{\alpha}) = E_{\theta} \left[(\alpha Z_2^r - \sigma^r)^2 \right]$$

is minimized at

$$\boxed{17/30}$$

$$\alpha \equiv \alpha(\underline{\theta}) = \sigma^r \frac{E_{\underline{\theta}}(Z_2^r)}{E_{\underline{\theta}}(Z_2^{2r})}$$

$$= \frac{E(G_{n-1}^r)}{E(G_{n-1}^{2r})} \quad (G_{n-1} \sim \text{Gamma}(1, n-1))$$

$$= \frac{\sqrt{n-1+r}}{\sqrt{n-1+2r}}, \quad r > -(n-1)$$

Which does not depend on $\underline{\theta}$ (i.e. for any $\underline{\theta} \in \Theta$, the minimiser $\alpha_0 = \frac{\sqrt{n-1+r}}{\sqrt{n-1+2r}}$ is the same).

Thus the minimum MSE estimator is

$$\hat{\sigma}_{\alpha_0} = \alpha_0 \left(\sum_{i=1}^n (X_i - x_{(1)})^r \right)^{1/r}, \quad \text{where } \alpha_0 = \frac{\sqrt{n-1+r}}{\sqrt{n-1+2r}}$$

For $\alpha \in \mathbb{R}$ and $\underline{\theta} \in \Theta$,

$$\begin{aligned} R(\underline{\theta}, \hat{\sigma}_{\alpha}) &= E_{\underline{\theta}} \left((\alpha Z_2^r - \sigma^r)^2 \right) \\ &= \alpha^2 E_{\underline{\theta}}(Z_2^{2r}) - 2\alpha \sigma^r E_{\underline{\theta}}(Z_2^r) + \sigma^{2r} \\ &= \alpha^2 \sigma^{2r} \frac{\sqrt{n-1+2r}}{\sqrt{n-1}} - 2\alpha \sigma^{2r} \frac{\sqrt{n-1+r}}{\sqrt{n-1}} + \sigma^{2r} \\ &= \left(\alpha^2 \frac{\sqrt{n-1+2r}}{\sqrt{n-1}} - 2\alpha \frac{\sqrt{n-1+r}}{\sqrt{n-1}} + 1 \right) \sigma^{2r} \end{aligned}$$

The UMVUE of σ^r is

$$\hat{\sigma}_{\alpha_1}(X) = \alpha_1 \left(\sum_{i=1}^n (X_i - x_{(1)})^r \right)^{1/r}$$

Where

$$\alpha_1 = \left(E(G_{n-1}^r) \right)^{-1} = \frac{\sqrt{n-1}}{\sqrt{n-1+r}}$$

(iii)

$$P_0(X_1 \leq s_p) = p \Rightarrow 1 - e^{-\frac{s_p - \mu}{\sigma}} = p$$

$$\Rightarrow s_p = \mu - \ln(1-p)\sigma$$

The UMVUE of s_p is

$$\begin{aligned} \hat{s}_p &= x_{(1)} - \frac{1}{n(n-1)} \sum_{i=1}^n (X_i - x_{(1)}) = \ln(1-p) \times \frac{1}{n-1} \sum_{i=1}^n (X_i - x_{(1)}) \\ &= x_{(1)} - \left(\frac{1}{n} + \ln(1-p) \right) S_{n-1} \end{aligned}$$

(IV) The value of $\psi_6(1)$ is

$$g_6(x) = P_0(x_1 \leq t_0 \mid x_{(1)}, \sum_{i=2}^n (x_i - x_{(1)})$$

For $x \in \mathbb{R}^n$, $n \geq 2$,

$$P_0(x_1 \leq t_0 \mid x_{(1)} = x, \sum_{i=2}^n (x_i - x_{(1)}) = \lambda)$$

$$= P_0\left(\frac{x_1 - x_{(1)}}{\sum_{i=2}^n (x_i - x_{(1)})} \leq \frac{t_0 - x}{\lambda} \mid x_{(1)} = x, \sum_{i=2}^n (x_i - x_{(1)}) = \lambda\right)$$

$$= P_0\left(\frac{x_1 - x_{(1)}}{\sum_{i=2}^n (x_i - x_{(1)})} \leq \frac{t_0 - x}{\lambda}\right) \quad (\text{Basu's Theorem})$$

Let $Z = \frac{x_1 - x_{(1)}}{\sum_{i=2}^n (x_i - x_{(1)})}$. Then Z is ancillary and λ_0

for finding distribution of Z , WLOG, we may assume

that $\mu = 0$ and $\sigma = 1$.

Note that $P_0(Z \geq 0) = 1$ and $P_0(Z = 0) = P_0(x_1 = x_{(1)}) = \frac{1}{n}$.

Finding distribution of Z is tedious.

(*) ←

Problem No. 7

$(\bar{X}, \bar{Y}, \sum_{i=1}^n (X_i - \bar{X})^2, \sum_{i=1}^n (Y_i - \bar{Y})^2)$ is a C-S Statistic

(Can be verified using result for exponential family)

$$\bar{X} - \bar{Y} \sim N(\mu_1 - \mu_2, \sigma^2), \quad \text{where } \sigma^2 = \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n}$$

We know that if $X \sim N(\mu, \sigma^2)$, $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$, $E(Z) = E(Z^3) = 0$

$$E(Z^2) = 1, \quad E(Z^4) = 3$$

$$\Rightarrow E(X) = \mu, \quad E(X^2) = \mu^2 + \sigma^2$$

$$E((X - \mu)^3) = 0 \Rightarrow E(X^3) = 3\mu(\mu^2 + \sigma^2) - 3\mu^3 + \mu^3$$

$$= \mu^3 + 3\mu\sigma^2$$

$$E((X - \mu)^4) = 3\sigma^4 \Rightarrow E(X^4) = 4\mu(\mu^3 + 3\mu\sigma^2) - 6\mu^2(\sigma^2 + \mu^2)$$

$$+ 4\mu^4 - \mu^4 + 3\sigma^4$$

$$= \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$$

[19/30]

$$E(\bar{X} - \bar{Y}) = \mu_1 - \mu_2; \quad E((\bar{X} - \bar{Y})^2) = (\mu_1 - \mu_2)^2 + \sigma^2$$

$$E((\bar{X} - \bar{Y})^3) = (\mu_1 - \mu_2)^3 + 3(\mu_1 - \mu_2)\sigma^2; \quad E((\bar{X} - \bar{Y})^4) = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$$

$= \mu^4 + 6(E(\bar{X}^2) - \sigma^2)\sigma^2 + 3\sigma^4$
 $= \mu^4 + 6E(\bar{X}^2)\sigma^2 - 3\sigma^4$

$$\text{Let } S_1^2 = \frac{1}{m-1} \sum_{i=1}^m (x_i - \bar{x})^2, \quad S_2^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$$

$$(m-1)S_1^2 \sim \sigma_1^2 \chi_{m-1}^2, \quad (n-1)S_2^2 \sim \sigma_2^2 \chi_{n-1}^2$$

Using independence of $\bar{x}, \bar{y}, \sum_{i=1}^m (x_i - \bar{x})^2$ and $\sum_{i=1}^n (y_i - \bar{y})^2$

$$E\left(\frac{S_1^2}{m} + \frac{S_2^2}{n}\right) = E(\hat{\sigma}^2) = \sigma^2$$

$$\left(\hat{\sigma}^2 = \frac{S_1^2}{m} + \frac{S_2^2}{n} \right) \sim \frac{\sigma_1^2 \chi_{m-1}^2}{m(m-1)} + \frac{\sigma_2^2 \chi_{n-1}^2}{n(n-1)}$$

$$\sigma^4 = \frac{(\sigma_1^2)^2}{m^2} + \frac{(\sigma_2^2)^2}{n^2} + 2 \frac{\sigma_1^2 \sigma_2^2}{m n}$$

$$E[(m-1)^2 (S_1^2)^2] = \sigma_1^4 E[(\chi_{m-1}^2)^2] = (m-1)(m+1)\sigma_1^4$$

$$E\left[\frac{m-1}{m+1} S_1^4\right] = \sigma_1^4$$

$$E\left[\frac{n-1}{n+1} S_2^4\right] = \sigma_2^4$$

$$\text{Let } \hat{\sigma}^4 = \frac{(m-1)}{m^2(m+1)} S_1^4 + \frac{(n-1)}{n^2(n+1)} S_2^4 + 2 \frac{S_1^2}{m} \frac{S_2^2}{n}$$

$$\text{Then } E(\hat{\sigma}^4) = \sigma^4$$

Now using independence of $\bar{x}, \bar{y}, \sum_{i=1}^m (x_i - \bar{x})^2$ and $\sum_{i=1}^n (y_i - \bar{y})^2$

UNQUE of $\psi_1(\underline{\theta}) = \mu_1 - \mu_2$ is $\delta_1(\underline{x}) = \bar{x} - \bar{y}$,

UNQUE of $\psi_2(\underline{\theta}) = (\mu_1 - \mu_2)^2$ is $\delta_2(\underline{x}) = (\bar{x} - \bar{y})^2 - \hat{\sigma}^2$

UNQUE of $\psi_3(\underline{\theta}) = (\mu_1 - \mu_2)^3$ is $\delta_3(\underline{x}) = (\bar{x} - \bar{y})^3 - 3(\bar{x} - \bar{y})\hat{\sigma}^2$

UNQUE of $\psi_4(\underline{\theta}) = (\mu_1 - \mu_2)^4$ is $\delta_4(\underline{x}) = (\bar{x} - \bar{y})^4 - 6\bar{x}^2\hat{\sigma}^2 + \hat{\sigma}^4$.

$$(11) \quad \begin{matrix} (m-1)S_1^2 \sim \sigma_1^2 \chi_{m-1}^2 \\ (n-1)S_2^2 \sim \sigma_2^2 \chi_{n-1}^2 \end{matrix} \quad \left. \vphantom{\begin{matrix} (m-1)S_1^2 \sim \sigma_1^2 \chi_{m-1}^2 \\ (n-1)S_2^2 \sim \sigma_2^2 \chi_{n-1}^2 \end{matrix}} \right\} \text{ independent}$$

$$(m-1)^{r/2} E(S_1^r) = \sigma_1^r E((X_{m-1}^L)^{r/2}) = \frac{2^{r/2} \frac{\Gamma(\frac{m-1+r}{2})}{\Gamma(\frac{m-1}{2})} \sigma_1^r}{\Gamma(\frac{m-1}{2})} = \frac{\sigma_1^r}{C_{0,m,r} \Gamma(r/2)}$$

$r > m-1$

$$(n-1)^{-1/2} E(S_2^{-1}) = \frac{\sigma_2^{-1}}{C_{0,n,-1}}$$

$$\Rightarrow \frac{(m-1)^{r/2}}{(n-1)^{1/2}} C_{0,m,r} C_{0,n,-1} E\left(\frac{S_1^r}{S_2}\right) = \frac{\sigma_1^r}{\sigma_2}$$

$$\Rightarrow g_5(\underline{x}) = \frac{(m-1)^{r/2}}{(n-1)^{1/2}} C_{0,m,r} C_{0,n,-1} \frac{S_1^r}{S_2} \text{ is the UMVUE of } \frac{\sigma_1^r}{\sigma_2}$$

(Pr1) The joint pdf of $\underline{X}, \underline{Y}$ is

$$h_0(\underline{x}, \underline{y}) = \frac{1}{\sqrt{k} \sigma_1} \times \frac{1}{\sigma_2 (2\pi)} e^{-\frac{1}{2k\sigma_1^2} \sum_{i=1}^m (x_i - \mu_1)^2 - \frac{1}{2\sigma_2^2} \sum_{i=1}^n (y_i - \mu_2)^2}$$

$$= \frac{1}{2\pi \sqrt{k} \sigma_1} e^{-\frac{1}{2k\sigma_1^2} \left[\sum_{i=1}^m (x_i - \bar{x})^2 + k \sum_{i=1}^n (y_i - \bar{y})^2 + m(\bar{x} - \mu_1)^2 + n k (\bar{y} - \mu_2)^2 \right]}$$

$$\Rightarrow T = (\bar{X}, \bar{Y}, \sum_{i=1}^m (x_i - \bar{x})^2 + k \sum_{i=1}^n (y_i - \bar{y})^2) \text{ is a CS Statistic}$$

$$E\left[\sum_{i=1}^m (x_i - \bar{x})^2 + k \sum_{i=1}^n (y_i - \bar{y})^2\right] = (m-1)\sigma_1^2 + k(n-1)\sigma_1^2 = (m+n-2)\sigma_1^2$$

$$Z = \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{m} + \frac{\sigma_1^2}{n}}} \sim N(0,1)$$

$$E(Z^3) = 0 \Rightarrow E((\bar{X} - \bar{Y})^3) - 3(\mu_1 - \mu_2) \left[\left(\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n} \right) + (\mu_1 - \mu_2)^2 \right] + 3(\mu_1 - \mu_2)^3 - (\mu_1 - \mu_2)^3 = 0$$

$$E[(\bar{X} - \bar{Y})^3] = 3(\mu_1 - \mu_2) \left(\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n} \right) + (\mu_1 - \mu_2)^3 = 3(\mu_1 - \mu_2) \left(\frac{1}{n} + \frac{1}{n} \right) \sigma_1^2 + (\mu_1 - \mu_2)^3$$

$$T = \sum_{i=1}^m (X_i - \bar{X})^2 + n \sum_{i=1}^n (Y_i - \bar{Y})^2 \sim \sigma_1^2 \chi_{m+n-2}^2$$

$$E(T^{3/2}) = \sigma_1^3 E(\chi_{m+n-2}^{3/2})$$

$$= 2^{3/2} \frac{\sqrt{\frac{m+n-1}{2}}}{\sqrt{\frac{m+n-2}{2}}} \sigma_1^3 = \frac{1}{\sqrt{m+n}} \sigma_1^3$$

The UNBUE of $\psi_7(\theta) = \sigma_1^2$ is

$$\delta_7(x) = \sqrt{m+n} T^{3/2}, \text{ where } \sqrt{m+n} = \frac{\sqrt{\frac{m+n-1}{2}}}{2^{3/2} \sqrt{\frac{m+n-2}{2}}}$$

UNBUE of $\psi_6(\theta) = (\mu_1 - \mu_2)^2$ (using independence of $\bar{X} - \bar{Y}$ and T)

$$\delta_6(x) = (\bar{X} - \bar{Y})^2 - 3(\bar{X} - \bar{Y}) \left(\frac{1}{n} + \frac{1}{n} \right) \frac{T}{m+n-2}$$

Problem No. 8 $T = (X_{(1)}, \sum_{i=1}^m (X_i - X_{(1)}), Y_{(1)}, \sum_{i=1}^n (Y_i - Y_{(1)}))$ is c.s

$$Z_1 = \frac{m(X_{(1)} - \mu_1)}{\sigma_1} \sim \text{Exp}(1)$$

$$Z_2 = \frac{n(Y_{(1)} - \mu_2)}{\sigma_2} \sim \text{Exp}(1)$$

$$Z_3 = \frac{\sum_{i=1}^m (X_i - X_{(1)})}{\sigma_1} \sim \text{Gamma}(m-1)$$

$$Z_4 = \frac{\sum_{i=1}^n (Y_i - Y_{(1)})}{\sigma_2} \sim \text{Gamma}(n-1)$$

} mutually independent
... (1)

22/30

$$E(X_{(1)}) = \mu_1 + \frac{\sigma_1}{m} E(Z_1) = \mu_1 + \frac{\sigma_1}{m}$$

$$E(Y_{(1)}) = \mu_2 + \frac{\sigma_2}{n}$$

$$E\left(\sum_{i=1}^m (X_i - X_{(1)})\right) = \sigma_1 E(Z_1) = (m-1)\sigma_1$$

$$E\left(\sum_{i=1}^n (Y_i - Y_{(1)})\right) = (n-1)\sigma_2$$

$$(1) \quad E\left(X_{(1)} - \underbrace{\frac{1}{m(m-1)} \sum_{i=1}^m (X_i - X_{(1)})}_{= \hat{\mu}_1 \text{ (a) }}\right) = \mu_1$$

$$E\left(Y_{(1)} - \underbrace{\frac{1}{n(n-1)} \sum_{i=1}^n (Y_i - Y_{(1)})}_{= \hat{\mu}_2}\right) = \mu_2$$

$$\Rightarrow E(\underbrace{\hat{\mu}_1 - \hat{\mu}_2}_{\text{b) of GS Statistic } T}) = \mu_1 - \mu_2$$

$$\Rightarrow \delta_1(x) = \hat{\mu}_1 - \hat{\mu}_2 \text{ is the UMVUE of } \mu_1 - \mu_2$$

Also want problem (c)

$$E\left(X_{(1)}^2 - \frac{2\sigma_1}{m} X_{(1)}\right) = \mu_1^2$$

$$E\left(Y_{(1)}^2 - \frac{2\sigma_2}{n} Y_{(1)}\right) = \mu_2^2$$

Using mutual independence described in (1)

$$E(\hat{\mu}_1 \hat{\mu}_2) = \mu_1 \mu_2$$

$$E\left(X_{(1)}^2 - \frac{2X_{(1)}}{m(m-1)} \sum_{i=1}^m (X_i - X_{(1)})\right) = \mu_1^2$$

$$E\left(Y_{(1)}^2 - \frac{2Y_{(1)}}{n(n-1)} \sum_{i=1}^n (Y_i - Y_{(1)})\right) = \mu_2^2$$

$$\text{Let } \hat{\sigma}_1^2 = \frac{1}{m-1} \sum_{i=1}^m (X_i - X_{(1)}) \quad \text{and} \quad \hat{\sigma}_2^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - Y_{(1)})$$

$$x_{(1)} = \hat{\mu}_1 + \frac{\hat{\sigma}_1}{m}, \quad y_{(1)} = \hat{\mu}_1 + \frac{\hat{\sigma}_1}{h}$$

$$E_0 \left(\hat{\mu}_1 + \frac{\hat{\sigma}_1^2}{m} + 2 \frac{\hat{\mu}_1 \hat{\sigma}_1}{m} - \frac{2}{m} \left(\hat{\mu}_1 + \frac{\hat{\sigma}_1}{m} \right) \hat{\sigma}_1 + \hat{\mu}_1^2 + \frac{\hat{\sigma}_1^2}{m} + 2 \frac{\mu_1 \sigma_1}{h} - \frac{2}{h} \left(\hat{\mu}_1 + \frac{\hat{\sigma}_1}{h} \right) \hat{\sigma}_1 - 2 \hat{\mu}_1 \hat{\mu}_1 \right) = (\mu_1 - \mu_2)^2$$

$$E_0 \left[\left(\hat{\mu}_1 - \hat{\mu}_2 \right)^2 - \frac{2\mu_1}{m} \hat{\sigma}_1 - \frac{2\mu_1}{h} \hat{\sigma}_1^2 \right] = (\mu_1 - \mu_2)^2$$

↳ UNBUE of $\psi_2(\theta) = (\mu_1 - \mu_2)^2$

$$(ii) \quad E_0 \left(\frac{\sum_{i=1}^m (x_i - x_{(1)})^2}{\sum_{i=1}^h (y_i - y_{(1)})^2} \right) = E_0 \left(\frac{\sigma_1^2 z_3^2}{\sigma_2^2 z_4^2} \right)$$

$$= \psi_3(\theta) \quad E(z_3^2) \times E\left(\frac{1}{z_4^2}\right) \quad \begin{matrix} \text{if } z_3 \sim N(0,1) \\ \text{and } z_4 \sim N(0,1) \end{matrix}$$

$$= \frac{\Gamma_{m-1+r} \Gamma_{h-1}}{\Gamma_{m-1} \Gamma_{h-1}}$$

$$\Rightarrow S_{3LY} = \frac{\Gamma_{m-1+r} \Gamma_{h-1}}{\Gamma_{m-1} \Gamma_{h-1}} \frac{\left(\sum_{i=1}^m (x_i - x_{(1)})^2 \right)^r}{\left(\sum_{i=1}^h (y_i - y_{(1)})^2 \right)^r} \quad \text{is the UNBUE of}$$

$$\psi_3(\theta) = \frac{\sigma_1^2}{\sigma_2^2}$$

(iii) An in Problem 7 (iii), it can be verified that

$$T = (x_{(1)}, y_{(1)}, \sum_{i=1}^m (x_i - x_{(1)}) + k \sum_{i=1}^h (y_i - y_{(1)}))$$

is CS.

Also by Basu's Theorem

$(x_{(1)}, y_{(1)})$ and $\sum_{i=1}^m (x_i - x_{(1)}) + k \sum_{i=1}^h (y_i - y_{(1)})$ are independent.

$$U = \sum_{i=1}^m (X_i - \bar{X}_{(1)}) + k \sum_{i=1}^n (Y_i - \bar{Y}_{(1)}) = \sigma_1 (Z_1 + Z_2) \sim \sigma_1 \text{ (equal, } m+n-2)$$

$$E[U] = \sigma_1 (m+n-2)$$

$$E\left[X_{(1)} - \frac{U}{m(m+n-2)}\right] = \mu_1$$

$$E\left[Y_{(1)} - \frac{U}{kn(m+n-2)}\right] = \mu_2$$

$$\text{Let } \hat{\mu}_1 = X_{(1)} - \frac{U}{m(m+n-2)}, \quad \hat{\mu}_2 = Y_{(1)} - \frac{U}{kn(m+n-2)}$$

$$\text{Let } \hat{\sigma}_1^2 = \frac{U}{m+n-2}$$

Then

$$E(\hat{\sigma}_1^2) = \sigma_1^2$$

$$E(\hat{\sigma}_1^4) = \frac{1}{(m+n-2)^2} E(U^2) = \frac{m+n-1}{m+n-2} \sigma_1^4$$

$$\Rightarrow E\left(\frac{m+n-2}{m+n-1} \hat{\sigma}_1^2\right) = \sigma_1^2$$

unbiased of σ_1^2

$$E(\hat{\mu}_1 - \hat{\mu}_2) = \mu_1 - \mu_2$$

$$E[(\hat{\mu}_1 - \hat{\mu}_2)^2] = \text{Var}(\hat{\mu}_1 - \hat{\mu}_2) + (\mu_1 - \mu_2)^2$$

$$= \text{Var}\left(X_{(1)} - Y_{(1)} - \left(\frac{1}{m} - \frac{1}{kn}\right) \hat{\sigma}_1^2\right)$$

$$= \frac{\sigma_1^2}{m^2} + \frac{\sigma_1^2}{k^2 n^2} + \left(\frac{1}{m} - \frac{1}{kn}\right)^2 * \frac{\sigma_1^4}{m+n-2}$$

$$+ (\mu_1 - \mu_2)^2$$

$$\Rightarrow E[(\hat{\mu}_1 - \hat{\mu}_2)^2] = \underbrace{\left(\frac{m+n-2}{(m+n-1)m^2} + \frac{m+n-2}{(m+n-1)k^2 n^2} - \frac{1}{m+n-1} * \left(\frac{1}{m} - \frac{1}{kn}\right)^2\right)}_{\text{unbiased of } \text{Var}(\hat{\mu}_1 - \hat{\mu}_2)} \hat{\sigma}_1^2 + (\mu_1 - \mu_2)^2$$

Problem 14.9

(i) $T_1 = \sum_{i=1}^m X_i$ is a c.s. statistic, with $T_1 \sim \text{Poisson}(m\theta)$
 $\psi_1(\theta) = P_{\theta_1}(X_1 \in D) = e^{-\theta_1} \sum_{j=0}^{\infty} \frac{\theta_1^{2j+1}}{(2j+1)!} = e^{-\theta_1} \left[\frac{e^{\theta_1} - e^{-\theta_1}}{2} \right] = \frac{1 - e^{-2\theta_1}}{2}$

The UNBUE of $\psi_1(\theta)$ is $\delta_1(T_1) = \frac{1}{2} - \frac{1}{2} \delta_1^*(T_1)$,

Where

$E_{\theta_1}(\delta_1^*(T_1)) = e^{-2\theta_1} \quad \forall \theta_1 > 0$
 $\Rightarrow \sum_{k=0}^{\infty} \delta_1^*(k) \frac{e^{-m\theta_1} (m\theta_1)^k}{k!} = e^{-2\theta_1} \quad \forall \theta_1 > 0$

$\Rightarrow \sum_{k=0}^{\infty} \frac{\delta_1^*(k) m^k}{k!} \theta_1^k = e^{(m-2)\theta_1} \quad \forall \theta_1 > 0$

$\Rightarrow \sum_{k=0}^{\infty} \frac{\delta_1^*(k) m^k}{k!} \theta_1^k = \sum_{k=0}^{\infty} \frac{(m-2)^k}{k!} \theta_1^k \quad \forall \theta_1 > 0$

$\rightarrow$ Two power series matched on an open interval $(0, \infty)$
 $\approx$ Coefficients of the two power series are the same

$\Rightarrow \frac{\delta_1^*(k) m^k}{k!} = \frac{(m-2)^k}{k!} \quad k=0, 1, 2, \dots$

$\Rightarrow \delta_1^*(T_1) = \left(1 - \frac{2}{m}\right)^{T_1}$

$\Rightarrow$ The UNBUE of $\psi_1(\theta) = P_{\theta_1}(X_1 \in D)$ is

$\delta_1(T_1) = \frac{1}{2} \left[1 - \left(1 - \frac{2}{m}\right)^{T_1} \right]$

(ii) $E_{\theta}((\bar{X} - \bar{Y})^2) = \text{Var}_{\theta}(\bar{X} - \bar{Y}) + (E_{\theta}(\bar{X} - \bar{Y}))^2$
 $= \frac{\theta_1}{m} + \frac{\theta_2}{n} + (\theta_1 - \theta_2)^2 \quad \forall \theta \in \Theta$

$\Rightarrow E_{\theta}((\bar{X} - \bar{Y})^2 - \frac{\bar{X}}{m} - \frac{\bar{Y}}{n}) = (\theta_1 - \theta_2)^2 \quad \forall \theta \in \Theta$

$$\Rightarrow S_2(\underline{x}; \underline{I}) = (\bar{x} - \bar{\tau})^2 - \frac{\bar{x}}{m} - \frac{\bar{\tau}}{n} \text{ is the UMVUE of } \psi_2(\theta) = (\theta_1 - \theta_2)^2$$

Problem No. 10

(i) See Problem 13 (a), Assignment No. 3

~~(ii) Note that $E_f(\bar{x}) = E_f(x_1)$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \sum_{i=1}^n x_{(i)}$ is a function of C-S Statistic $\underline{I} = (x_{(1)}, \dots, x_{(n)})$. Thus $\bar{x}$ is the UMVUE of $E_f(x_1)$.~~

(iii) Note that $E_f(\bar{x}) = E_f(x_1)$, $\forall f \in \mathcal{P}$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \sum_{i=1}^n x_{(i)}$ is a function of C-S Statistic $\underline{I} = (x_{(1)}, \dots, x_{(n)})$. Thus $\bar{x}$ is the UMVUE of $E_f(x_1)$.

$$\text{Let } S_{n-1}^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \left[\sum_{i=1}^n x_{(i)}^2 - n\bar{x}^2 \right].$$

Then $E_f(S_{n-1}^2) = \text{Var}_f(x_1)$, $\forall f \in \mathcal{P}$ and S_{n-1}^2 is a function of C-S Statistic $\underline{I} = (x_{(1)}, \dots, x_{(n)})$.

$\Rightarrow S_{n-1}^2$ is the UMVUE of $\sigma^2 = \text{Var}_f(x_1)$.

$$\gamma_1 = \theta^2 = (E_f(x_1))^2 = (E_f(x_1^2) - \text{Var}_f(x_1))$$

$$= E_f \left[\frac{1}{n} \sum_{i=1}^n x_i^2 - S_{n-1}^2 \right]$$

$\hookrightarrow$ function of C-S Statistic $\underline{I} = (x_{(1)}, \dots, x_{(n)})$

$$\hat{\gamma}_1 = \frac{1}{n} \sum_{i=1}^n x_i^2 - S_{n-1}^2 = -\frac{1}{n(n-1)} \sum_{i=1}^n x_{(i)}^2 + \frac{n}{n-1} \bar{x}^2 \text{ is the UMVUE of } \gamma_1 = \theta^2 = \frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} x_i x_j$$

Alt. way: $E_f \left(\frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} x_i x_j \right) = \theta^2$, $\forall f \in \mathcal{P}$ and $\frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} x_i x_j$ is a function of C-S Statistic $\underline{I} = (x_{(1)}, \dots, x_{(n)})$

The UMVUE of $\gamma_2 = P_f(X_1 \leq t_0)$ is

$$\hat{\gamma}_2 = P_f(X_1 \leq t_0 | (X_{(1)}, \dots, X_{(n)}))$$

Since $(X_1, \dots, X_n) \stackrel{d}{=} (X_{a_1}, \dots, X_{a_n})$ for any permutation $(a_1, \dots, a_n)$ of $(1, \dots, n)$ (as $X_1, \dots, X_n$ are i.i.d.), we have

~~$$P_f(X_1 = X_{(1)}) = P_f(X_2 = X_{(1)})$$~~

$$P_f(X_1 = X_{(1)} | (X_{(1)}, \dots, X_{(n)})) = P_f(X_2 = X_{(1)} | (X_{(1)}, \dots, X_{(n)}))$$

$$\dots = P_f(X_n = X_{(1)} | (X_{(1)}, \dots, X_{(n)})) = \frac{1}{n}, \quad i=1, \dots, n$$

Thus the UMVUE of $\gamma_2 = P_f(X_1 \leq t_0)$ is

$$\hat{\gamma}_2 = P_f(X_1 \leq t_0 | (X_{(1)}, \dots, X_{(n)}))$$

$$= \begin{cases} 0, & t_0 < X_{(1)} \\ \frac{1}{n}, & X_{(1)} \leq t_0 < X_{(2)} \\ \frac{2}{n}, & X_{(2)} \leq t_0 < X_{(3)} \\ \vdots \\ \frac{i}{n}, & X_{(i)} \leq t_0 < X_{(i+1)}, \quad i=1, \dots, n-1 \\ \frac{n-1}{n}, & X_{(n-1)} \leq t_0 < X_{(n)} \\ 1, & t_0 \geq X_{(n)} \end{cases}$$

$$t_0 < X_{(1)}$$

$$X_{(1)} \leq t_0 < X_{(2)}$$

$$X_{(2)} \leq t_0 < X_{(3)}$$

$$X_{(i)} \leq t_0 < X_{(i+1)}, \quad i=1, \dots, n-1$$

$$X_{(n-1)} \leq t_0 < X_{(n)}$$

$$t_0 \geq X_{(n)}$$

→ Empirical d.f.

Problem No. 11

(1) Let θ be the point of symmetry. Then

$$(X_1 - \theta, \dots, X_n - \theta) \stackrel{d}{=} (\theta - X_1, \dots, \theta - X_n)$$

$$\Rightarrow (X_{(1)} - \theta, \dots, X_{(n)} - \theta) \stackrel{d}{=} (\theta - X_{(1)}, \dots, \theta - X_{(n)})$$

~~$$\Rightarrow P_\theta(X_{(1)} - \theta > X_{(n)} - \theta) = P(\theta - X_{(1)} > \theta - X_{(n)})$$~~

$$\Rightarrow P_b \left(\frac{x_{(1)} - \theta + x_{(n)} - \theta}{2} > \frac{x_{(1)} - \theta + \dots + x_{(n)} - \theta}{n} \right) \\ = P_b \left(\frac{\theta - x_{(n)} + \theta - x_{(1)}}{2} > \frac{\theta - x_{(n)} + \dots + \theta - x_{(1)}}{n} \right), \quad \forall b \in \theta$$

$$\Rightarrow P_b \left(\frac{x_{(1)} + x_{(n)}}{2} > \frac{x_{(1)} + \dots + x_{(n)}}{n} \right) \\ = P_b \left(\frac{x_{(1)} + x_{(n)}}{2} < \frac{x_{(1)} + \dots + x_{(n)}}{n} \right), \quad \forall b \in \theta$$

$$\Rightarrow P_b \left(\frac{x_{(1)} + x_{(n)}}{2} > \frac{x_{(1)} + \dots + x_{(n)}}{n} \right) = \frac{1}{2}, \quad \forall b \in \theta$$

$$\text{Let } h(x_{(1)}, \dots, x_{(n)}) = \begin{cases} \frac{1}{2}, & \text{if } \frac{x_{(1)} + x_{(n)}}{2} > \frac{x_{(1)} + \dots + x_{(n)}}{n} \\ -\frac{1}{2}, & \text{if } \frac{x_{(1)} + x_{(n)}}{2} \leq \frac{x_{(1)} + \dots + x_{(n)}}{n} \end{cases}$$

$$\text{Then } E_b [h(x_{(1)}, \dots, x_{(n)})] = \frac{1}{2} [2 P_b \left(\frac{x_{(1)} + x_{(n)}}{2} > \frac{x_{(1)} + \dots + x_{(n)}}{n} \right) - 1] \\ = 0, \quad \forall b \in \theta$$

$$\text{But } P_b (h(x_{(1)}, \dots, x_{(n)}) = 0) = 0, \quad \forall b \in \theta.$$

$\Rightarrow T(X) = (x_{(1)}, \dots, x_{(n)})$ is not complete.

(ii) On Contrary Suppose that the UMVUE θ_0 exist.
Let $P_1 = \{N(\theta, 1) : \theta \in \mathbb{R}\} \subseteq \mathcal{P}$. $\bar{X}$ is UMVUE under P_1 .

$$\Rightarrow \theta_0 = \bar{X} \quad \text{a.n. } b \in P_1$$

$$\Rightarrow \theta_0 = \bar{X} \quad \text{a.n. with respect to Lebesgue measure} \\ (\text{Since Lebesgue measure is dominated by any } b \text{ in } P_1)$$

$$\text{Let } P_2 = \{U(\theta_1 - \theta_2, \theta_1 + \theta_2) : \theta_1 \in \mathbb{R}, \theta_2 > 0\}$$

29/20

Then $\frac{x_{(1)} + x_{(n)}}{2}$ is the ^{unique} UMVUE under P_2

$$\Rightarrow \bar{x} = \frac{x_{(1)} + x_{(n)}}{2} \quad \text{a.s. } b \in P_2$$

$\rightarrow$ Contradiction

(iii) Let $P_1 = \{N(0,1) : 0 \in \mathbb{R}\}$. Then x_1 is complete under $b \in \mathcal{P}$. Then

$$E_b[h(x_1)] = 0, \quad \forall b \in \mathcal{P} \Rightarrow P_b[h(x_1) = 0] = 1, \quad \forall b \in \mathcal{P}$$

$$\Rightarrow h(x_1) = 0, \quad \text{a.e. Lebesgue measure}$$

$\Rightarrow \delta_0(x) = 0$ is the unique unbiased estimator of 0 under $\mathcal{P}$

Since $E(x_1) = \mu \in \mathbb{R}$, x_1 is the unique UMVUE of μ .